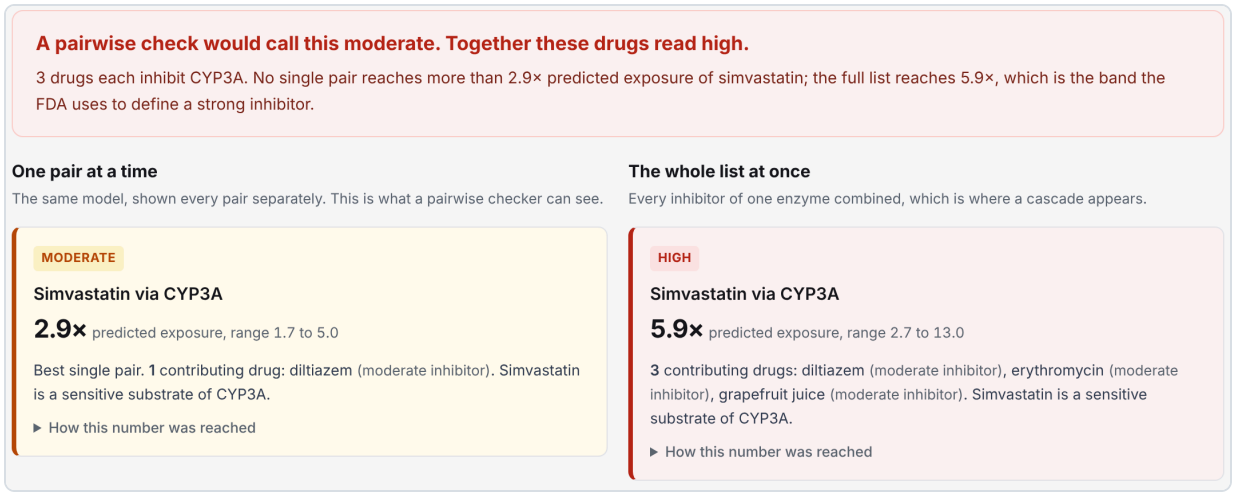
Cascade

The drug interaction a pairwise check cannot see.

Interaction checkers compare drugs two at a time. A patient on seven drugs has 21 pairs, and every one of them can be cleared while three of those drugs each partly inhibit the same enzyme, because the combined effect exists in no pair. Cascade scores the whole list at once, with a model built only from the FDA's published interaction classes, and shows the pairwise view beside it.



Simvastatin, diltiazem, erythromycin, grapefruit juice, metoprolol, omeprazole and warfarin. Best single pair: 2.9× predicted exposure, moderate. The three CYP3A inhibitors together: 5.9×, high. Drop any one of them and the finding disappears.

Across the FDA's index table of 205 drugs

Combination	Count	Understated by a pairwise view
2 inhibitors of one enzyme + 1 substrate	84,302	8,057 (9.6%)
3 inhibitors of one enzyme + 1 substrate	1,341,070	78,753 (5.9%)

"Understated": the full list lands in a higher severity band than any single pair of the same drugs. Of random 7-drug lists from the same table, 3.1% hold at least one (62 of 2,000).

How it works

1. **Roles**: 285 published roles for 205 drugs, taken from the FDA's index table. A drug not on it is reported as not covered, never guessed.
2. **Intervals**: every FDA class is a range (a moderate inhibitor raises exposure 2 to 5×), and the ranges are carried through the arithmetic. The fraction metabolised is derived from the class definitions, not assumed.
3. **Combine**: inhibitors of one enzyme add competitively, $A = 1/(1+\Sigma r)$, $AUCR = 1/(fm \cdot A + 1 - fm)$.
4. **Compare**: the baseline is the same model run one pair at a time, so the only difference between the two views is the cascade.

Measured

Enumeration of every 2- and 3-inhibitor combination in the table	1.4 million, nothing sampled
Calibration: one inhibitor reads back in its own FDA class	weak, moderate and strong all hold
Common shortcut it replaces (multiplying inhibitions), three moderate inhibitors	up to 125× against 13× competitive
Tests, including the worked example and the calibration	19 passing

Limits, stated up front

- It predicts exposure, not harm. Dose, kidney and liver function, genotype and timing are not modelled.
- One enzyme at a time, reversible inhibition only: no transporters, no QT or serotonergic stacking.
- Only the 205 drugs on the FDA's index list; the FDA calls the list a guide, not a complete one.
- Not validated against clinical outcomes. It shows what a pairwise view misses, not that a missed cascade harmed anyone.

Educational prototype, not clinical decision support. Data: U.S. FDA, Drug Development and Drug Interactions, Table of Substrates, Inhibitors and Inducers, Tables 1 and 2, retrieved 31 Oct 2026. Built by Joshua Anojulu with Claude (Anthropic), which wrote the code; every number comes from evaluate.py and the tests in the repository.